

① What is Regularization in Machine Learning?
why is it needed?

→ Regularization is a technique used to reduce overfitting in machine learning models.

- Sometimes a model learns the training data too perfectly, including noise and unnecessary patterns. Because of this, it performs poorly on new unseen data.
- Regularization helps the model become simpler & more generalized.

→ without regularization:

- Model becomes too complex.
- Training accuracy becomes very high
- Testing accuracy becomes low
- Model memorizes data instead of learning patterns

This problem is called overfitting.

Ex: Imagine a student:

Memorizing answers → Overfitting.

Understanding Concepts → Good generalization.

Regularization forces the ML model to focus on important features only.

Normal loss function:

$$\text{Loss} = \text{Error}$$

Regularized loss function:

$$\text{Loss} = \text{Error} + \text{Penalty}$$

② Difference between Ridge (L_2) & Lasso (L_1) Reg.

Feature	Ridge (L_2)	Lasso (L_1)
Penalty Type	Square of coefficients	Absolute values of coefficients
Formula	$\lambda \sum w^2$	$\lambda \sum w $
Remove Features	No	Yes
Coefficients	Become small	Some become exactly zero
Best Use	When all features are important.	When feature selection is needed
Overfitting Reduction	Yes	Yes

→ Ridge Regression (L_2) :

Add square of weights as Penalty.

$$\text{Loss} = \text{Error} + \lambda \sum w^2$$

Effect :
 Reduces coefficient values
 keeps all features
 Prevents overfitting.

Ex: Suppose features are : Age, salary, Height, weight.

- Ridge reduces importance but keeps all features.

→ Lasso regression (L_1):

- Adds absolute value of weights as penalty.

$$\text{Loss} = \text{Error} + \lambda \sum |w|$$

Effect: Some coefficients become zero
Automatically removes unimportant features
Performing features selection.

Ex: If "Height" is not useful, Lasso may completely remove it.

③ What is cross-validation and why is it important?

→ Cross-validation is a technique used to check how well a machine learning model performs on unseen data.

- Instead of training and testing only once, the dataset is divided multiple times for better evaluation.
- It helps us know: Is the model accurate?
Is the model overfitting?
Will it work on real-world data?

- Better model evaluation.
- Reduces overfitting.
- Efficient use of data.
- Helps in model selection.

④ cross-validation Techniques:

[A] K-fold cross-validation:

- Dataset is divided into K equal parts.

Ex: $K=5$.

Dataset split into 5 parts.

Process: Use 4 folds for training.
Use 1 fold for testing.
Repeat 5 times.

Final Accuracy = Average of all accuracies.

→ Advantages: Better accuracy estimate.
Efficient use of data.
widely used.

[B] Stratified K-fold cross-validation:

- Used mainly for classification problems.
- It keeps the same class ratio in every fold.

Ex: Suppose dataset has: 80% cats
20% dogs.

Stratified K-fold ensures every fold also has:
80% cats
20% dogs.

→ without stratification:

Some folds may have very few samples of one class.

model evaluation becomes biased.

→ Best used when Dataset is imbalanced.

1c Leave - One - Out - cross - validation (LOOCV):

→ Special type of K-Fold where:
 $K = \text{number of samples}$

If dataset has 100 rows:

Train on 99 rows

Test on 1 row

Repeat 100 times

Advantages: Uses maximum data for training
very accurate evaluation

Disadvantages: very slow for large datasets.
computationally expensive.

1d Time Series Split:

→ Used for time-based data like:

Stock prices
Weather data
Sales forecasting

- In time series, future data should never be used to predict past data.

→ why special method needed?

- Random splitting breaks time order.
- Time series split keeps chronological order.

⑤ why are Tree-Based models Less Sensitive to Feature Scaling?

→ Tree-based models: Decision Tree
Random Forest
XGBoost

do not depend on distances or magnitudes

They split data using conditions like:

Age > 25

or

Salary < 50000

So scaling is unnecessary.

→ why Scaling Matters in other Algorithms

Algorithms like: KNN

SVM

Linear Regression

depend on: Distance

Gradient calculations

- So scaling becomes important there.